Type MS · Explain how enzymes work in biosensors and test strips

Answer 1 of 5 · Question 6 part b.ii · 4 marks

2020 NOV · Paper 01 · extract 1 of 2 · find part b.ii in this mark scheme extract

6 (a)	(i)	sucrase/invertase (1)		1
	(ii)	• enzyme is a protein (1) • biuret test (1) • purple if protein present/remains blue if absent (1)		3
	(iii)	• reusable (1) • stable (1)		2
(b)	(i)	• allows glucose molecules to pass (1) • prevents larger/other molecules/ blood cells from passing (1)		2
	(ii)	• breaks down/converts glucose (1) • into hydrogen peroxide (1) • which activates electrode (1) • more glucose results in more hydrogen peroxide (1)		4

Total 12 marks

Type MS · Explain how enzymes work in biosensors and test strips

Answer 1 of 5 · Question 6 part b.ii · 4 marks

2020 NOV · Paper 01 · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how enzymes work in biosensors and test strips

Answer 2 of 5 · Question 6 part b.i · 2 marks

2020 NOV · Paper 01 · extract 1 of 2 · find part b.i in this mark scheme extract

6 (a)	(i)	sucrase/invertase (1)		1
	(ii)	• enzyme is a protein (1) • biuret test (1) • purple if protein present/remains blue if absent (1)		3
	(iii)	• reusable (1) • stable (1)		2
(b)	(i)	• allows glucose molecules to pass (1) • prevents larger/other molecules/ blood cells from passing (1)		2
	(ii)	• breaks down/converts glucose (1) • into hydrogen peroxide (1) • which activates electrode (1) • more glucose results in more hydrogen peroxide (1)		4

Total 12 marks

Type MS · Explain how enzymes work in biosensors and test strips

Answer 2 of 5 · Question 6 part b.i · 2 marks

2020 NOV · Paper 01 · extract 2 of 2 · find part b.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how enzymes work in biosensors and test strips

Answer 3 of 5 · Question 3 part a.ii · 2 marks

2021 JUN · Paper 02 · extract 1 of 2 · find part a.ii in this mark scheme extract

number				
3	(a) (i)	water;		1
	(ii)	• allows urea to pass; • prevents other larger molecules from passing;	Allow small molecules go through	2
	(iii)	• breaks down urea; • to produce ammonium ions; • to react with/stimulate sensor;	Allow ammonia Allow idea of being picked up/detected by sensor	3
	(b)	• enzyme activity increases/enzyme works faster/higher rate of reaction; • at higher temperatures; • more ammonium ions produced; • higher reading;	Ignore details related to enzyme denaturing / collision theory/kinetic energy ORA for lower temperature	4
	(c)	any two from • can be reused; • products not contaminated; • enzymes more stable; • cost effective justified;	Allow can withstand temperature/pH changes	2

Total 12 marks

Type MS · Explain how enzymes work in biosensors and test strips

Answer 3 of 5 · Question 3 part a.ii · 2 marks

2021 JUN · Paper 02 · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how enzymes work in biosensors and test strips

Answer 4 of 5 · Question 3 part a.iii · 3 marks

2021 JUN · Paper 02 · extract 1 of 2 · find part a.iii in this mark scheme extract

number				
3	(a) (i)	water;		1
	(ii)	• allows urea to pass; • prevents other larger molecules from passing;	Allow small molecules go through	2
	(iii)	• breaks down urea; • to produce ammonium ions; • to react with/stimulate sensor;	Allow ammonia Allow idea of being picked up/detected by sensor	3
	(b)	• enzyme activity increases/enzyme works faster/higher rate of reaction; • at higher temperatures; • more ammonium ions produced; • higher reading;	Ignore details related to enzyme denaturing / collision theory/kinetic energy ORA for lower temperature	4
	(c)	any two from • can be reused; • products not contaminated; • enzymes more stable; • cost effective justified;	Allow can withstand temperature/pH changes	2

Total 12 marks

Type MS · Explain how enzymes work in biosensors and test strips

Answer 4 of 5 · Question 3 part a.iii · 3 marks

2021 JUN · Paper 02 · extract 2 of 2 · find part a.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how enzymes work in biosensors and test strips

Answer 5 of 5 · Question 7 part b · 2 marks

2025 JUN · Paper 02 (regional R) · find part b in this mark scheme extract

7	(a)	amylase;		1
	(b)	Any two from • enzymes are immobilised on testing strip; • react with/breaks down glucose (in urine); • (catalyse a chemical reaction that) causes a colour change (in strip);		2
	(c)	pancreas;		1
	(d)	• rejection/immune response/allergies; • (immune system) recognised it as foreign/antigen;		2
	(e)	any two from • use restriction enzymes; • cut out insulin gene from human DNA; • insert human insulin gene into yeast genome/plasmid/genetic material;		2
			Total for Question 7 = 8 marks	